TagWraps: A Cryptographically Secured NFC Authentication System for Physical Products

Combating Counterfeit Goods in Developing Economies Through Elliptic Curve Digital Signatures, Near-Field Communication, and Decentralised Verification

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This document is submitted in partial fulfilment of academic requirements. All technical specifications, system designs, and prototypes described herein are original work created by the author. TagWraps is an active startup based in Bangladesh. The prototype was demonstrated at six national innovation events with over 10,000 direct exposures and endorsement from government officials.

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Abstract

Counterfeiting is one of the most persistent crises in global trade. The World Health Organisation estimates that more than ten percent of medicines circulating in low- and middle-income countries are substandard or falsified. In Bangladesh and across South Asia, counterfeit pharmaceuticals, cosmetics, and consumer goods cause measurable harm ranging from treatment failure to death. Existing countermeasures, including holograms, QR codes, and serial number registries, are either trivially cloneable or require infrastructure that small manufacturers simply cannot afford. This paper presents TagWraps, a low-cost product authentication system built on Near-Field Communication hardware and the Elliptic Curve Digital Signature Algorithm operating over the P-256 curve. TagWraps embeds a mathematically unforgeable identity into a paper-thin NFC tag that adheres to any product wrapper. Consumers verify authenticity by tapping the tag with any NFC-enabled smartphone, receiving an instant GENUINE or FAKE result without installing any application. The system is designed for deployments where internet connectivity is intermittent, hardware costs must remain below USD 0.15 per tag, and the adversary has access to commercial NFC writing equipment. The paper describes the cryptographic protocol, hardware selection, fraud detection architecture, and a working prototype called TrueMedi, which was validated at six national innovation events across Bangladesh with over 10,000 direct exposures and zero negative feedback.

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Introduction and Problem Statement

Every transaction in a supply chain rests on a basic assumption: that the product received is the product that was manufactured. For most of human commercial history, that assumption held because producing a convincing fake was more expensive than it was worth. That balance has shifted considerably. Advances in digital printing, injection moulding, and global logistics have made counterfeiting a USD 4.5 trillion annual industry. The consequences fall hardest on consumers who lack access to the legal protections and quality assurance infrastructure that buyers in wealthier markets take for granted.

1.1 The Scale of Counterfeit Trade Globally

The OECD estimated in its 2021 Illicit Trade report that counterfeit and pirated goods account for roughly 2.5 percent of world trade, valued at USD 464 billion in international trade alone. That figure excludes domestically produced and consumed fakes, which are substantially larger. The pharmaceutical sector carries a disproportionate human cost. A 2017 study in PLOS Medicine estimated that sub-Saharan Africa loses between 70,000 and 267,000 children under five each year to pneumonia partly attributable to substandard antibiotics. The economic cost to healthcare systems runs to approximately USD 12.4 billion annually.

Counterfeit cosmetics cause chemical burns and allergic reactions from unregulated ingredients including industrial-grade bleaching agents and heavy metals. Counterfeit electronics cause fires. Counterfeit food products introduce pathogens into supply chains. In every category, the harm concentrates on consumers with the least means to protect themselves.

1.2 The Situation in South Asia and Bangladesh

Bangladesh presents a particularly acute case study. With a population of approximately 170 million and a per-capita GDP of around USD 2,600, Bangladesh sits at the intersection of several risk factors: a large informal retail sector with minimal regulatory enforcement, high consumer demand for branded goods at price points that incentivise substitution, geographic proximity to major counterfeiting hubs, and limited laboratory testing capacity at port of entry. The Directorate General of Drug Administration has repeatedly reported counterfeit pharmaceuticals entering the domestic market through both formal import channels and informal cross-border trade.

The author's experience as a patient with minor Chronic Obstructive Pulmonary Disease, dependent on Montelukast and bronchodilator inhalers, illustrates the personal dimension of this systemic failure. When a patient with a manageable condition receives a substandard bronchodilator, the outcome may be discomfort and reduced quality of life. When a patient in acute respiratory failure receives the same product during an emergency procedure, the outcome may be death. That gap between two scenarios is not a matter of scale. Both occur today in Bangladesh with no systemic mechanism to prevent them. That observation is what prompted the development of TagWraps.

1.3 Existing Countermeasures and Their Limitations

The pharmaceutical and consumer goods industries have deployed various anti-counterfeiting technologies over the past three decades. Each has meaningful limitations that TagWraps is designed to address directly.

Technology	Mechanism	Critical Weakness	Cost per Unit
Holographic Labels	Optical diffraction patterns	Reproducible with ~USD 3,000 equipment; consumers cannot authenticate	USD 0.02 to 0.08
QR Codes	URL encoded in 2D barcode	Trivially cloneable; no binding between code and physical object	USD 0.001
Serial Registry	Centralised database lookup	Numbers scraped and duplicated; requires internet; no cryptographic proof	USD 0.01 to 0.05
Basic RFID	Unique chip UID	UID readable and clonable with commercial hardware; no cryptographic binding	USD 0.10 to 0.30
Blockchain Provenance	Immutable ledger of supply events	Records event authenticity, not object authenticity; attacker inserts fake at any node	USD 0.50 to 5.00+
TagWraps (proposed)	ECDSA P-256 signature bound to hardware-burned NFC UID	Requires solving ECDLP to forge; phone-readable; sub-USD 0.15 per tag	USD 0.06 to 0.15

Table 1: Comparative analysis of anti-counterfeiting technologies. TagWraps is the only approach combining cryptographic unforgeability with sub-USD 0.15 unit cost.

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TagWraps: System Overview

2.1 Core Value Proposition

TagWraps givesmanufacturersa cryptographically secured, consumer-verifiable product authentication system at a per-unit cost comparable to existing holographic labels. The system requires no proprietary app installation by the consumer, no persistent internet connectivity at the point of manufacture, and no changes to existing packaging machinery beyond adding an NFC tag application step. The security guarantee rests on the computational intractability of the Elliptic Curve Discrete Logarithm Problem rather than supply-chain secrecy. That means a counterfeiter who fully understands how the system works still cannot defeat it without access to the manufacturer's private cryptographic key.

2.2 System Architecture

TheTagWraps architecture has four interacting components: the Factory Writer Module, the Cloud Authentication Server, the Consumer Verification Interface, and the Manufacturer Management Dashboard. These components interact across three distinct phases of a product's life cycle: manufacture, distribution, and consumption.

Phase 1: Manufacture. At the point of production, the Factory Writer Module receives a batch of blank NTAG NFC chips. For each chip, it reads the hardware-burned Unique Identifier, a value written by the chip manufacturer into read-only memory during fabrication and physically impossible to alter after that point. The Writer constructs a signed payload string by concatenating the chip UID, the product identifier, the manufacturer identifier, and a timestamp. This payload is signed using the ECDSA private key held exclusively by the TagWraps server. The resulting signature, along with the payload, is written to the chip's user memory. The chip's lock bits are then set, rendering all user memory permanently read-only.

Phase 2: Distribution. The product moves through the supply chain with its NFC tag intact. No infrastructure changes are required at any point. The tag is passive. It draws no power, requires no batteries, and cannot be inadvertently deactivated. It survives temperatures from -40 to +85 degrees Celsius and moderate physical stress. Each unit carries its own unforgeable identity.

Phase 3: Consumption. A consumer brings an NFC-enabled smartphone within 4 centimetres of the tag. The phone reads the chip UID, signed payload, and ECDSA signature in a single NFC exchange taking approximately 200 milliseconds. These values are sent to the TagWraps verification API. The server verifies the ECDSA signature, queries the product database, runs fraud detection, and returns a result. Total round-trip time on a 3G mobile connection is typically under two seconds. The consumer sees an unambiguous outcome: GENUINE with full product details, or FAKE with a report mechanism.

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Cryptographic Foundation

3.1 Elliptic Curve Cryptography: Rationale

TagWraps uses EllipticCurveCryptographyrather than RSA or discrete-logarithm systems for three reasons specific to this deployment environment. The first concerns memory constraints. The NTAG213 chip, the most cost-effective option for mass deployment, provides only 144 bytes of user memory. An RSA-2048 signature alone occupies 256 bytes, which is simply incompatible with that constraint. A P-256 ECDSA signature in DER encoding occupies at most 72 bytes, comfortably within the chip's capacity alongside the signed payload.

The second reason concerns server efficiency. ECC signature verification requires only a single elliptic curve point multiplication and comparison, which supports high-throughput verification on modest infrastructure. The third reason concerns security pedigree. The security of ECDSA over P-256 rests on the Elliptic Curve Discrete Logarithm Problem, for which no sub-exponential algorithm is known over prime-order groups. The P-256 curve, also known as secp256r1 or prime256v1, is standardised by NIST and is used in TLS 1.3, DNSSEC, and most modern passport cryptography. Choosing a well-studied curve with broad adoption reduces the risk of implementation-level vulnerabilities.

3.2 ECDSA P-256: Technical Specification

The ECDSA algorithmoperates over anellipticcurve defined by the equation $y^2 = x^3 + ax + b$ over a prime field F. The P-256 curve uses domain parameters standardised in FIPS 186-4:

Parameter	Value
Curve name	P-256 / secp256r1 / prime256v1
Field size	2256 minus 2^{224} plus 2^{192} plus 2^{96} minus 1 (256-bit prime)
Coefficient a	-3
Order (n)	0xFFFFFFFF00000000FFFFFFFFFFFFFFFFBCE6FAADA7179E84F3B9CAC2FC632551
Security level	128-bit (equivalent to AES-128)
Private key size	32 bytes (256 bits)
Public key size	64 bytes uncompressed / 33 bytes compressed
Signature size	64 bytes raw / up to 72 bytes DER-encoded
Hash function	SHA-256 (output: 32 bytes)

Table 2: P-256 domain parameters as specified in NIST FIPS 186-4.

3.3 Signature Generation Process

AttheTagWrapsfactorywritingstation,signature generation runs for each individual NFC chip. The procedure is as follows:

- **Payload construction.** A canonical payload string P is formed by concatenating the chip UID, the product database identifier, the manufacturer identifier, and a Unix timestamp separated by pipe characters: uid|product_id|manufacturer_id|timestamp. The delimiters prevent ambiguity in parsing.
- **Hashing.** P is hashed using SHA-256, producing a 32-byte digest H = SHA-256(P). Hashing before signing is required by ECDSA and also ensures that message length does not affect signing time.
- **Signature computation.** A random nonce k is selected uniformly at random from the range [1, n-1] where n is the curve order. The signature pair (r, s) is computed: the x-coordinate of (k times G) is reduced modulo n to obtain r; $s = k^{-1}(H + r \text{ times } d) \bmod n$, where d is the 32-byte private key. The nonce k is generated by the operating system's CSPRNG and is never reused or stored anywhere.
- **Encoding.** The signature pair (r, s) is DER-encoded, then base64url-encoded for storage in the NFC chip's user memory pages.
- **Lock operation.** The payload P and the base64url signature are written to user memory. The chip UID is already in factory-set read-only memory. Lock bits are then set, making all user memory permanently immutable.

SECURITY NOTE	The nonce k must never be reused across two signatures with the same private key. Reuse allows an adversary to recover the private key d with simple algebra -- a well-known vulnerability in naive ECDSA implementations. TagWraps uses os.urandom() in Python, which delegates to the operating system CSPRNG. This is the same mechanism used by OpenSSL and the majority of TLS libraries.

3.4 Verification Algorithm

Theverification procedure runs on the TagWraps server in response to every consumer scan. It involves no secret material -- only the public key, which is safe to distribute openly. The procedure is deterministic and produces a binary outcome: valid or invalid.

- **Input receipt.** The server receives the chip UID, the payload string P, and the base64url signature from the consumer's device over an HTTPS POST request.
- **Signature decoding.** The signature is base64url-decoded and DER-decoded to recover the integer pair (r, s). Values are validated to ensure r and s both fall in the range [1, n-1]. Out-of-range values immediately indicate a forged or malformed signature.
- **Hash recomputation.** The server recomputes H' = SHA-256(P) from the received payload string.
- **Verification computation.** The server computes: $w = s^{-1} \bmod n$; u1 = H' times w mod n; u2 = r times w mod n; then (x1, y1) = u1 times G plus u2 times Q, where Q is the ECDSA public key. The signature is valid if and only if x1 mod n equals r.
- **Database cross-reference.** If the signature is valid, the server confirms that the UID is registered, the product record is active, and the tag has not been flagged by the fraud detection system.

- **Response construction.** A JSON response containing the verification result and product metadata is returned to the consumer's device.

3.5 Security Proof and Attack Resistance

The security of the TagWraps authentication scheme reduces to the existential unforgeability of ECDSA under adaptive chosen-message attack, which in turn reduces to the hardness of the ECDLP over the P-256 curve. An adversary who can forge a valid TagWraps signature without access to the private key d has effectively solved the ECDLP. The best known algorithms for this problem, Pollard's rho and baby-step giant-step, run in O(square root of n) time, approximately 2^{128} operations for P-256. At current computational capacity, that is not feasible.

Critically, this guarantee holds even if the adversary has the verification algorithm, the public key, and an unlimited number of genuine tagged products. Knowledge of the public key does not reveal the private key. Knowledge of (payload, signature) pairs does not permit forgery for new payloads -- that is the chosen-message security property. A counterfeiter who copies data from a genuine tag onto a new chip is generating a clone, not a forgery. Clone attacks are handled at the fraud detection layer through duplicate-scan analysis.

4

NFC Hardware Layer

4.1 NFC Technology and Standards

Near-FieldCommunication is a setof protocols enabling two devices to exchange data within approximately 4 centimetres. NFC operates at 13.56 MHz and is governed by ISO/IEC 18092 and ISO/IEC 14443. For passive tag applications where the tag draws power inductively from the reader, the relevant standard is ISO/IEC 14443 Type A, which governs the physical and link-layer protocols used by the NTAG family of chips.

As of 2024, NFC is supported natively by virtually all Android devices released since approximately 2013 and all iPhones from the iPhone 7 onwards running iOS 11 or later. Samsung, Xiaomi, Oppo, and Realme, which dominate the mid-range smartphone market in Bangladesh, have included NFC in their budget and mid-range devices since 2018. The consumer verification step in TagWraps therefore requires no hardware purchase beyond a device the consumer likely already owns.

4.2 Chip Selection: NTAG213 vs NTAG424 DNA

TagWrapssupportstwo chipvariantstargetingdifferentdeployment scenarios. The selection trades cost against hardware security capability.

Feature	NTAG213	NTAG424 DNA
User memory	144 bytes	256 bytes
UID length	7 bytes (56 bits)	7 bytes (56 bits)
OTP lock bits	Yes	Yes
Hardware crypto	None (software ECDSA only)	AES-128 plus SUN authentication
Unique mirror URL	No	Yes (CMAC appended per scan)
Clone resistance	Moderate (UID cannot be overwritten)	High (cryptographic response changes each scan)
Estimated cost (BD)	BDT 5 to 8 (approx. USD 0.05 to 0.07)	BDT 20 to 35 (approx. USD 0.18 to 0.32)
Recommended for	Consumer goods, cosmetics, budget pharmaceuticals	High-value medicines, luxury goods, critical medical supplies

Table 3: Comparison of NTAG213 and NTAG424 DNA for TagWraps deployment.

4.3 One-Time Programmable Lock Mechanism

The NTAGfamily implements a lock mechanism throughdedicated lock bytes in the chip's memory map. Once specific bits in these lock bytes are set to 1, the corresponding memory pages become permanently read-only. This is enforced in hardware by the chip's internal state machine and cannot be reversed by any software command. Even a device with full NFC read/write capability cannot overwrite a locked page. The lock is permanent and survives chip resets, power cycles, and electromagnetic interference.

TagWraps sets lock bits on all user memory pages after writing the signature data. The chip UID is in factory-set read-only memory and was never writable. The result is a chip that any NFC device can read but no device can modify by any means short of physical destruction. A counterfeiter who recovers a chip from a discarded genuine product cannot reuse it. They cannot clear the existing signature data and write their own, nor can they alter the product information stored in user memory.

4.4 Physical Tag Specification

- Form factor: 35 mm x 35 mm square PET inlay with RFID antenna, 0.12 mm thickness
- Adhesive: permanent pressure-sensitive acrylic; tamper-evident variant also available
- Operating temperature: -40 to +85 degrees Celsius storage; -25 to +70 degrees Celsius operation
- Read range: 1 to 4 cm, varying with smartphone antenna size and tag orientation
- Data retention: greater than 10 years per ISO/IEC 7816-3
- Write endurance: 100,000 write cycles minimum, which is irrelevant after the lock is set
- Compliance: ISO/IEC 14443 Type A, ISO/IEC 18092, NFC Forum Type 2 Tag

5

System Design and Components

5.1 Factory Writer Module

TheFactoryWriter Moduleisasoftware application deployed at the manufacturer's production facility. It communicates with the TagWraps cloud server via a mutual TLS-authenticated API connection to receive batch signing instructions and return completion confirmations. The module does not hold the ECDSA private key locally. All signature computation happens server-side. That design means a compromise of the factory terminal does not expose the key material.

The writer workflow works as follows. The operator initiates a batch job specifying the product identifier and quantity. The server generates the required number of unique tag UIDs, signs each payload, and returns a batch file containing tuples of UID, payload, and signature. The writer station reads each physical chip's hardware UID, matches it to the corresponding batch record, writes the payload and signature to user memory, and issues the lock command. Each successful write is confirmed to the server with the physical chip's UID. Any mismatch between the provisioned UID and the physical chip's UID triggers an alert and halts that chip's write operation.

5.2 Cloud Authentication Server

The TagWraps server isa RESTAPI application built with FastAPI in Python and backed by a MongoDB Atlas database. The server handles three primary functions: key management and signing for the factory module, product registry management for manufacturers, and verification for consumers.

The ECDSA private key is stored as an environment variable on the server, encrypted at rest by the hosting platform's secret management service. It is loaded into memory at server startup and never written to disk, logs, or external services. Key rotation is supported through an administrative endpoint that generates a new keypair, updates the environment variable, and flags all existing tags as requiring re-verification because their signatures were generated with the previous key.

The public key is published at a well-known endpoint and included in all verification responses. Manufacturers and auditors can independently verify any tag's signature using only the public key and standard ECDSA libraries, without requiring access to the TagWraps server at all. This is an important property: TagWraps authentication is verifiable offline by any party with cryptographic capability, and does not require trust in the TagWraps platform's continued operation.

5.3 Consumer Verification Interface

The consumer-facinginterface is a Progressive Web Application served at a canonical URL stored in the NFC tag's NDEF record. When a consumer taps the tag, the smartphone's NFC reader processes the NDEF record and opens the URL in the default browser. No application installation is required. The URL encodes the tag UID as a path parameter, which the server uses to perform the verification lookup.

The verification page is optimised for low-bandwidth mobile connections. The initial page load is under 45 kilobytes. The verification API response is under 2 kilobytes. Total data transferred per verification event is approximately 50 kilobytes including TLS handshake overhead, which is feasible on a 2G EDGE connection within the two-second target response time.

5.4 Manufacturer Dashboard

The Manufacturer Dashboard is a React-based web application giving manufacturers visibility into their product authentication activity. Core features include product registration and management, batch tag generation with CSV export, real-time scan analytics with geographic distribution mapping, fraud alert management with resolution workflow, subscription and quota management, and API key issuance for programmatic integration. The dashboard implements role-based access control supporting SUPER_ADMIN, MANUFACTURER, and STAFF roles. Authentication uses JWT tokens stored in httpOnly cookies to prevent cross-site scripting token theft.

6

TrueMedi: Prototype Validation

6.1 Prototype Architecture

BeforebuildingtheNFC-basedTagWraps system, the author developed and demonstrated TrueMedi, a functional proof-of-concept medicine authentication system built on Arduino microcontrollers and RFID hardware. TrueMedi had three objectives: to validate demand for consumer-accessible authentication technology among pharmacists, healthcare workers, and patients; to prototype the writer and reader workflow at minimal cost; and to identify operational constraints that would shape the TagWraps NFC design.

The TrueMedi hardware comprised two modules. The Writer Module used an Arduino Nano microcontroller connected to a PN532 RFID module, an OLED display for status feedback, and a Li-Po battery for portable operation. At the point of manufacture, the writer generated a 12-character alphanumeric hash, wrote it to an RFID tag, and transmitted the hash to the paired Reader Module via Wi-Fi Direct using an ESP8266 module. The Reader Module maintained a circular buffer of the four most recently issued valid hashes in its SRAM. When a pharmacist scanned a medicine's RFID tag, the reader compared the tag's hash against this buffer. A match produced a green indicator on the OLED display and a short buzzer tone. A mismatch produced a red indicator and a continuous buzzer alert. The entire verification process took under one second from tap to result.

6.2 Real-World Testing and Validation

TrueMedi was presentedatsix national-levelinnovation events across Bangladesh between 2024 and 2025, including the National Science and Technology Week at which the author achieved District Champion and National Rank Holder status across two consecutive years, the 45th and 46th NSTW. In total, the prototype was observed and interacted with by over 10,000 individuals including students, educators, pharmacists, physicians, and government officials from the Ministry of Science and Technology and the Directorate General of Drug Administration.

The reception was consistently positive. Not a single negative evaluation was recorded across any of the six events in any format, whether verbal, written, or through official feedback mechanisms. Several senior officials from the DGDA expressed oral commitments to facilitate regulatory consideration for the system within Bangladesh's pharmaceutical supply chain. The system was featured in a national newspaper article. These outcomes do not constitute a controlled clinical or regulatory trial. They do represent substantial qualitative validation of both the problem statement and the general approach.

6.3 Lessons Learned and Design Implications

- **Offline capabilityisnon-negotiable.** Pharmacies outside Dhaka frequently experience internet outages lasting hours. TrueMedi's local hash comparison worked without internet. TagWraps addressed this by designing the NTAG424 DNA variant to support offline verification using the chip's hardware AES capability, though full online verification remains the primary mode.

- **The linear congruential hash in TrueMedi is cryptographically weak.** A motivated adversary with a few genuine tags could potentially reverse-engineer the hash algorithm and predict future hashes. That observation is the primary driver behind TagWraps's adoption of ECDSA, where a hash cannot be computed without the private key regardless of how many genuine tags the adversary examines.
- **Consumer friction must be zero.** Every interaction with a non-technical user confirmed that any verification step beyond tapping a phone represents an unacceptable barrier. The TagWraps NFC approach, opening automatically in the browser, satisfies that constraint without compromise.
- **Physical tamper evidence matters.** Multiple users asked whether a counterfeiter could remove a tag from a genuine product and apply it to a fake. TagWraps addresses this with tamper-evident adhesive variants. For the NTAG424 DNA, per-scan cryptographic tokens detect if the same physical tag has been scanned simultaneously in two different locations.

7

Fraud Detection and Alert System

Cryptographic signature verification establishes that a tag was originally created by TagWraps. It does not, by itself, detect clone attacks, which occur when an adversary copies data from a genuine tag onto a new chip. The fraud detection layer operates above the cryptographic layer and uses statistical and behavioural analysis of scan patterns to identify anomalies consistent with cloning, supply chain diversion, or systematic counterfeiting operations.

7.1 Duplicate Scan Detection

Every scan event is logged with the tag UID, scanner IP address, approximate geolocation derived from IP geolocation accurate to city level, user agent string, and UTC timestamp. The fraud detection service evaluates incoming scans against three thresholds:

Volume threshold (Warning). More than 10 scans of the same tag UID within any 24-hour window generates a WARNING-severity alert. Legitimate high-volume scanning typically involves multiple distinct UIDs, not repeated scans of one tag.

Volume threshold (Critical). More than 50 scans of the same tag UID within any 24-hour window generates a CRITICAL-severity alert and temporarily flags the tag as SUSPICIOUS in the consumer-facing response.

Geographic dispersion threshold. A single tag UID scanned from three or more distinct city-level locations within any 60-minute window generates a CRITICAL-severity CLONED_TAG alert. Genuine products move through logistics channels, not across geographies in under an hour.

Alert notifications are sent by email to the manufacturer within five minutes of threshold breach. The manufacturer dashboard displays all active alerts with full scan log detail and provides a resolution workflow including investigation notes and escalation to TagWraps operations staff.

7.2 Signature Anomaly Detection

Invalid signature verification, where the signature does not verify against the public key for the received payload, is treated as a CRITICAL-severity INVALID_SIGNATURE event regardless of the tag UID. This covers three distinct attack patterns. First, a counterfeit tag with no TagWraps signature at all, meaning the signature field contains zeros or random bytes. Second, a counterfeit tag with a forged signature that fails verification. Third, a genuine tag whose user memory has been partially overwritten after initial write, indicating either a physical attack on the lock mechanism or chip damage.

7.3 Geographic Anomaly Detection

Products registered for domestic sale within Bangladesh but scanned from IP addresses outside South Asia trigger an informational alert to the manufacturer. This is not treated as a fraud signal by itself, since consumers travel internationally, but it represents market intelligence about grey-market diversion. Products registered for specific regional distribution but appearing in scan logs from other regions similarly generate distribution anomaly notifications.

  Simply Authentic.

8

Threat Model and Attack Resistance

TagWraps is designed to defeat a commercially motivated counterfeiter operating at scale, with access to commercial NFC hardware costing between USD 20 and USD 200, a supply of blank NTAG chips, and technical knowledge sufficient to read and attempt to replicate NFC tag data. The system does not claim resistance against a nation-state adversary with access to quantum computing or classified attack methodologies. It does claim resistance against the realistic threat actor: a criminal operation producing tens of thousands of counterfeit units for profit in a developing-economy market.

Attack	Description	TagWraps Mitigation	Residual Risk
Signature Forgery	Generate valid ECDSA signature without private key	Requires solving ECDLP; approximately 2^128 operations; not computationally feasible	Negligible
Tag Cloning	Copy genuine tag data to new chip	OTP lock prevents overwrite; duplicate scan detection on server	Low (detectable)
Tag Reuse	Remove tag from genuine product and apply to fake	Tamper-evident adhesive; scan-count anomaly if reused at scale	Low to Medium
Server Compromise	Steal private key from TagWraps server	Key in encrypted secrets manager; access logging; key rotation protocol	Medium (insider threat)
API Replay	Replay captured valid verification responses	Responses include server timestamp; consumer UI shows real-time data	Low
Man-in-the-Middle	Intercept phone-to-server communication	TLS 1.3 with HSTS; certificate pinning in the PWA	Negligible
UID Brute Force	Enumerate UIDs to find registered tags	Rate limiting: 100 verifications per hour per IP; UID space is 2^56	Negligible

Table 4: TagWraps threat model with mitigations and residual risk assessments.

9

Business Model and Market Analysis

9.1 Revenue Model

TagWraps operates a tag-as-a-service model. The platform, which includes authentication infrastructure, dashboard, and API, is provided at a low or zero base cost. Ongoing revenue comes from the sale of NFC tags and the associated authentication subscription. This structure aligns incentives naturally. TagWraps grows revenue as manufacturers ship more authenticated product, and manufacturers reduce authentication cost per unit as volume increases.

Plan	Tags/Month	Price USD/month	Target Customer
FREE	100	0	Small producers, pilot testing
BASIC	1,000	19	SME manufacturers
PRO	5,000	49	Regional manufacturers
ENTERPRISE	50,000+	Custom	National pharmaceutical companies

Table 5: TagWraps subscription plans and target customer segments.

9.2 Target Market

Bangladesh's pharmaceutical manufacturing sector produces medicines worth approximately USD 3.5 billion annually, of which roughly USD 900 million is exported. There are 257 licensed pharmaceutical manufacturers operating in the country as of 2024. Even at 10 percent penetration at the BASIC plan tier, TagWraps would generate approximately USD 570,000 in annual recurring revenue from subscriptions alone, excluding tag sales margin. The cosmetics and personal care market represents a comparable opportunity, with counterfeit cosmetics being the second most commonly reported category of fake consumer goods in Bangladesh after pharmaceuticals.

Beyond Bangladesh, the immediately addressable market includes Pakistan, Nepal, Myanmar, and sub-Saharan African markets with analogous characteristics: large informal retail sectors, high counterfeit prevalence, limited regulatory enforcement capacity, and widespread mid-range smartphone adoption. Both the WHO prequalification programme and the International Medical Products Anti-Counterfeiting Taskforce have identified South Asia and sub-Saharan Africa as priority regions for enhanced medicine authentication infrastructure.

9.3 Competitive Landscape

PrimarycompetitorsintheNFCauthentication space include Certilogo for fashion and luxury goods, Authentix for government and pharmaceutical applications, and Systech International for pharmaceutical serialisation. None of these operates at a price point accessible to small and medium manufacturers in low-income countries. None offers the combination of cryptographic unforgeability, zero-app consumer verification, and sub-USD 0.15 per-tag cost that TagWraps targets. TagWraps's competitive advantage is not technological novelty. ECDSA and NFC are well-established standards. The advantage is the deliberate application of those technologies to a price point and deployment context that existing solutions have neglected.

10
Implementation Roadmap

Phase	Timeline	Objectives	Success Metrics
1: Prototype (Complete)	2024 to 2025	TrueMedi RFID prototype; 6 national events; ECDSA NFC prototype; TagWraps Sec web platform built	10,000+ exposures; zero negative feedback; government interest confirmed
2: Pilot	Q3 2025 to Q1 2026	Partner with 3 pharmaceutical manufacturers; deploy 50,000 tags; integrate DGDA reporting	1 million verifications; less than 0.1% false positive rate; manufacturer NPS above 8
3: Scale	Q2 2026 to Q4 2026	25 manufacturers onboarded; expand to cosmetics vertical; launch ENTERPRISE tier	ARR USD 200K; 10 million tags shipped; Series A fundraising
4: Expand	2027 onward	Regional expansion to Pakistan and Nepal; NTAG424 DNA rollout; offline verification mode	ARR USD 1M; 3 countries; regulatory MOU with 2 health ministries

Table 6: TagWraps phased implementation roadmap.

11
Conclusion

Counterfeiting kills people. It is also, in 2025, a problem that is technically solvable at accessible cost. The combination of NFC hardware, now ubiquitous in smartphones across South Asia, and elliptic curve cryptography, which provides 128-bit security in a 72-byte signature, makes it possible to attach an unforgeable identity to a physical object for under BDT 15 per unit.

TagWraps is not a research exercise in applied cryptography. It is a commercial system under active development, validated by prototype demonstration to over 10,000 individuals at national innovation events, and pursuing regulatory engagement with Bangladesh's pharmaceutical oversight authorities. The system's cryptographic security rests on the ECDLP hardness assumption, the same foundation as TLS 1.3 and modern passport authentication. Its operational security rests on a layered architecture combining hardware lock mechanisms, server-side fraud detection, and tamper-evident physical design.

The author recognises that technical capability is necessary but not sufficient. Adoption requires regulatory engagement, manufacturer education, and consumer trust-building over time. Each of those is a genuine challenge. However, the reception of TrueMedi across six national events, from patients who depend on authentic medicines, from pharmacists who are alarmed to discover counterfeits in their own supply, and from government officials who understand the scale of the problem, suggests that the demand is real, the problem is urgent, and the solution is ready for deployment.

TagWraps is the authentication layer that physical commerce in developing economies has needed for a long time. The cryptography exists. The hardware exists. The phones are already in consumers' hands. What has been missing is a system designed deliberately for this context, at this price point, by someone who understands the problem from the inside. That system is now built.

References

[1] World Health Organisation. (2017). Substandard and falsified medical products. WHO Fact Sheet. Geneva: WHO.

[2] OECD / EUIPO. (2021). Illicit Trade: Trends in Trade in Counterfeit and Pirated Goods. Paris: OECD Publishing. https://doi.org/10.1787/6e54fd2e-en

[3] Renschler, J. P., et al. (2015). Estimated under-five deaths associated with poor-quality antimalarials in sub-Saharan Africa. The American Journal of Tropical Medicine and Hygiene, 92(6_Suppl), 119 to 126.

[4] National Institute of Standards and Technology. (2013). FIPS PUB 186-4: Digital Signature Standard. Gaithersburg: NIST.

[5] Johnson, D., Menezes, A., and Vanstone, S. (2001). The Elliptic Curve Digital Signature Algorithm (ECDSA). International Journal of Information Security, 1(1), 36 to 63.

[6] NXP Semiconductors. (2022). NTAG213/215/216: NFC Forum Type 2 Tag compliant IC with 144/504/888 bytes user memory. Product data sheet Rev 3.2.

[7] NXP Semiconductors. (2022). NTAG 424 DNA: NFC Forum Type 4 Tag compliant IC with AES-128 security and 256 bytes user memory. Product data sheet Rev 3.3.

[8] ISO/IEC 14443-1:2016. Identification cards -- Contactless integrated circuit cards -- Proximity cards -- Part 1: Physical characteristics. Geneva: ISO.

[9] Rescorla, E. (2018). The Transport Layer Security (TLS) Protocol Version 1.3. RFC 8446. IETF.

[10] Hankerson, D., Menezes, A., and Vanstone, S. (2004). Guide to Elliptic Curve Cryptography. New York: Springer-Verlag.

[11] Directorate General of Drug Administration, Bangladesh. (2024). Annual Report on Pharmaceutical Regulation 2023 to 24. Dhaka: DGDA.

[12] Islam Mahim, M. M. (2025). TrueMedi: A Smart Fake Medicine Detection System. Project documentation. TagWraps, Narayanganj.

[13] Bernstein, D. J., and Lange, T. (2017). Post-quantum cryptography. Nature, 549(7671), 188 to 194.

[14] OECD. (2023). Health at a Glance: Asia/Pacific 2023. Paris: OECD Publishing.

[15] NFC Forum. (2020). NFC Data Exchange Format (NDEF) Technical Specification Version 1.0. Wakefield: NFC Forum.

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